

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Answers:

1. Concept Map

Application



Presentation



Session



Transport

(Segmentation, Error Recovery)



Network

(Routing, IP Address)



Data Link

(Framing, MAC Address, Error Detection)



Physical

(Bits Transmission)



Reliable Data Transmission

Explanation:

- Application Layer → Provides network services to users (HTTP, FTP, Email).
- Presentation Layer → Performs data encryption, compression, and translation.
- Session Layer → Establishes, manages, and terminates communication sessions.
- Transport Layer → Ensures reliable delivery through segmentation, flow control, and error recovery.
- Network Layer → Routes packets using IP addresses to the destination.
- Data Link Layer → Creates frames, uses MAC addresses, and detects transmission errors.
- Physical Layer → Transmits data as bits through cables or wireless media.

Analysis:

Each OSI layer performs a specific task and passes the processed data to the next layer. Together, the layers provide reliable, secure, and efficient communication from the sender to the receiver. This layered approach makes troubleshooting easier and improves network performance.

2. Concept Map of Data Units

Application



Presentation



Session



Transport



Segment



Network



Packet



Data Link



Frame



Physical



Bits



Transmission

Explanation:

- Transport Layer: Divides data into Segments.
- Network Layer: Adds IP address to form Packets.
- Data Link Layer: Adds MAC address and trailer to form Frames.
- Physical Layer: Converts frames into Bits for transmission.

Analysis:

- Each layer changes the data format to support reliable communication.
- This process is called encapsulation, making transmission accurate and efficient.
- Different data units help each layer perform its specific function effectively.
- Encapsulation allows data to travel securely and accurately through the network.
- It also enables communication between different networking devices and technologies.
- This improves the overall reliability of data transmission.

3. Concept for data link

Data Link Layer



MAC Address



Local Delivery

Network Layer



IP Address



Global Delivery

Local Delivery



Global Delivery



Complete End-to-End Communication

Explanation:

- The **Data Link layer** is responsible for communication within a local network (LAN).
- It uses the **MAC (Media Access Control) address**, which is a unique physical address assigned to every network interface card.
- MAC addresses ensure that frames are delivered correctly between devices connected to the same network.

- The **Network layer** uses **IP addresses** to identify devices across different networks.
- Routers use IP addresses to determine the best route for forwarding packets from the source network to the destination network.
- This enables communication over the Internet and other wide-area networks.

Analysis:

- Both MAC and IP addresses are essential for complete communication.
- MAC addresses handle local delivery between neighboring devices, while IP addresses enable communication across multiple interconnected networks.
- Together, they provide efficient, reliable, and accurate end-to-end data delivery.
- Without MAC addressing, local communication would fail, and without IP addressing, communication beyond the local network would not be possible.

4.Error Control

Error Control



Data Link Layer



Error Detection
(CRC, Checksum)



Transport Layer



Error Correction
(Acknowledgement,
Retransmission)



Reliable Communication

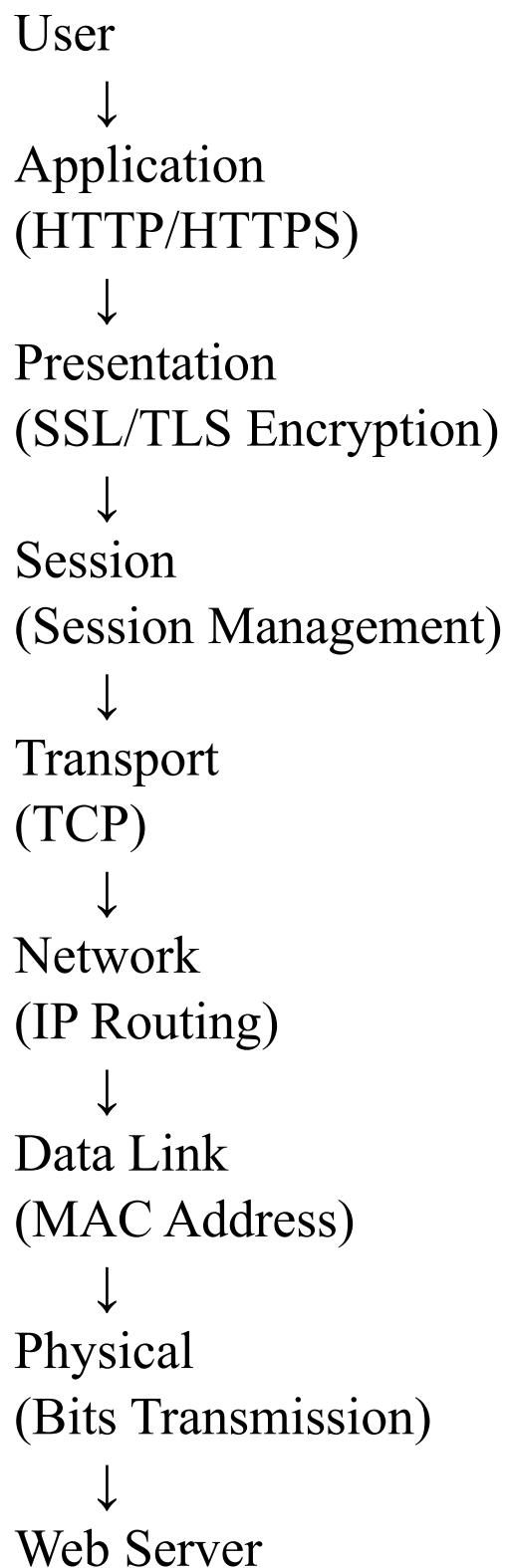
Explanation:

- Error control is an important feature that ensures the receiver obtains the same data sent by the sender.
- The Data Link layer detects transmission errors using techniques such as Cyclic Redundancy Check (CRC) and checksum.
- These methods determine whether data has been corrupted during transmission.
- The Transport layer, especially when using the TCP protocol, provides error correction by acknowledging received segments and retransmitting lost or corrupted data.
- It also ensures that data arrives in the correct order and prevents duplicate data from being delivered.

Analysis:

- The combined operation of the Data Link and Transport layers significantly improves network reliability.
- Error detection alone cannot recover lost information, while error correction ensures damaged or missing data is retransmitted.
- These mechanisms reduce packet loss, improve communication quality, and ensure that important information reaches its destination accurately even in noisy or unreliable network environments.
- Error detection and correction reduce data loss during transmission.
- They ensure that corrupted data is identified and replaced with the correct information.
- This increases network performance and provides accurate communication between sender and receiver.

5. Develop a concept map for web browsing using OSI layers



Explanation:

- When a user opens a website, the request passes through all seven OSI layers.
- The Application layer generates an HTTP or HTTPS request.
- The Presentation layer encrypts the data using SSL/TLS for secure communication.
- The Session layer establishes and manages the communication session between the browser and the server.
- The Transport layer uses TCP to divide the request into segments and guarantees reliable delivery.
- The Network layer routes packets using IP addresses.
- The Data Link layer creates frames using MAC addresses for local delivery.
- Finally, the Physical layer transmits the bits through wired or wireless media to the destination web server.

Analysis:

- When a user opens a website, the request passes through all seven OSI layers.
- The Application layer generates an HTTP or HTTPS request. The Presentation layer encrypts the data using SSL/TLS for secure communication.
- The Session layer establishes and manages the communication session between the browser and the server.
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6. Analyse the network issue:

Physical ✓



Data Link ✓



Network ✗



Transport ✗



Application ✗

Problem Begins



Network Layer

(IP Address / Routing Error)

Explanation:

- In this network issue, the Physical and Data Link layers are working correctly, but the Network layer has failed.
- This may be due to an incorrect IP address, gateway, or routing problem.
- Because of this failure, the Transport and Application layers cannot communicate properly.

Analysis:

- The concept map shows that the problem starts at the Network layer. Checking IP configuration, routing tables, and gateway settings can resolve the issue. Identifying the faulty layer reduces troubleshooting time and restores normal network communication.